NASA SPACE APPS 2026 AUTONOMOUS DEEP-SPACE FLIGHT DECK

21 SUITES PASSED (124 TESTS)

# Sol Check — End-to-End System Architecture, Clinical Workflow & Diagram Suite

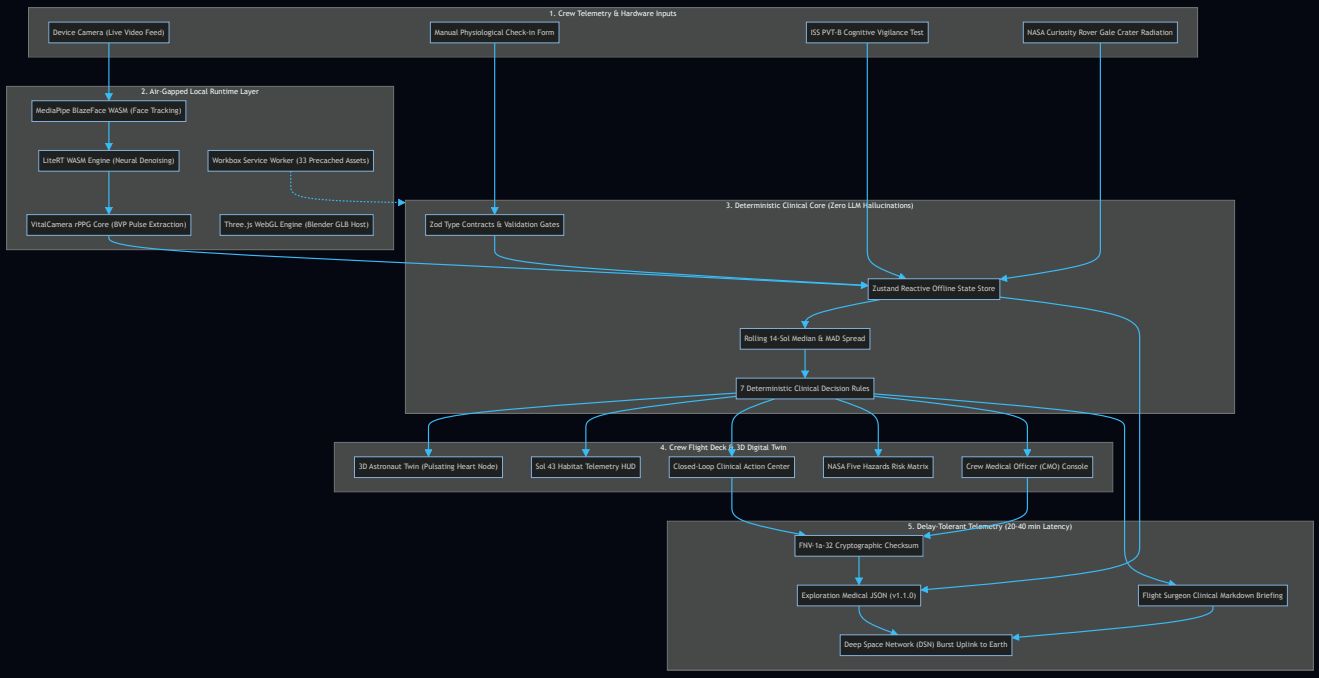
Complete operational architecture manual for autonomous crew health evaluation on Mars Sol 43 under 20–40 minute round-trip interplanetary communication latency.

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> **Operational Mission Context:** At an average orbital distance of 225 million kilometers from Earth, electromagnetic signals face a 20-minute one-way delay (40 minutes round-trip). In acute clinical anomalies, post-solar particle events, or insidious physiological deconditioning, astronauts cannot wait for Mission Control on Earth. Clinical intelligence must be autonomous, explainable, and air-gapped on-device.

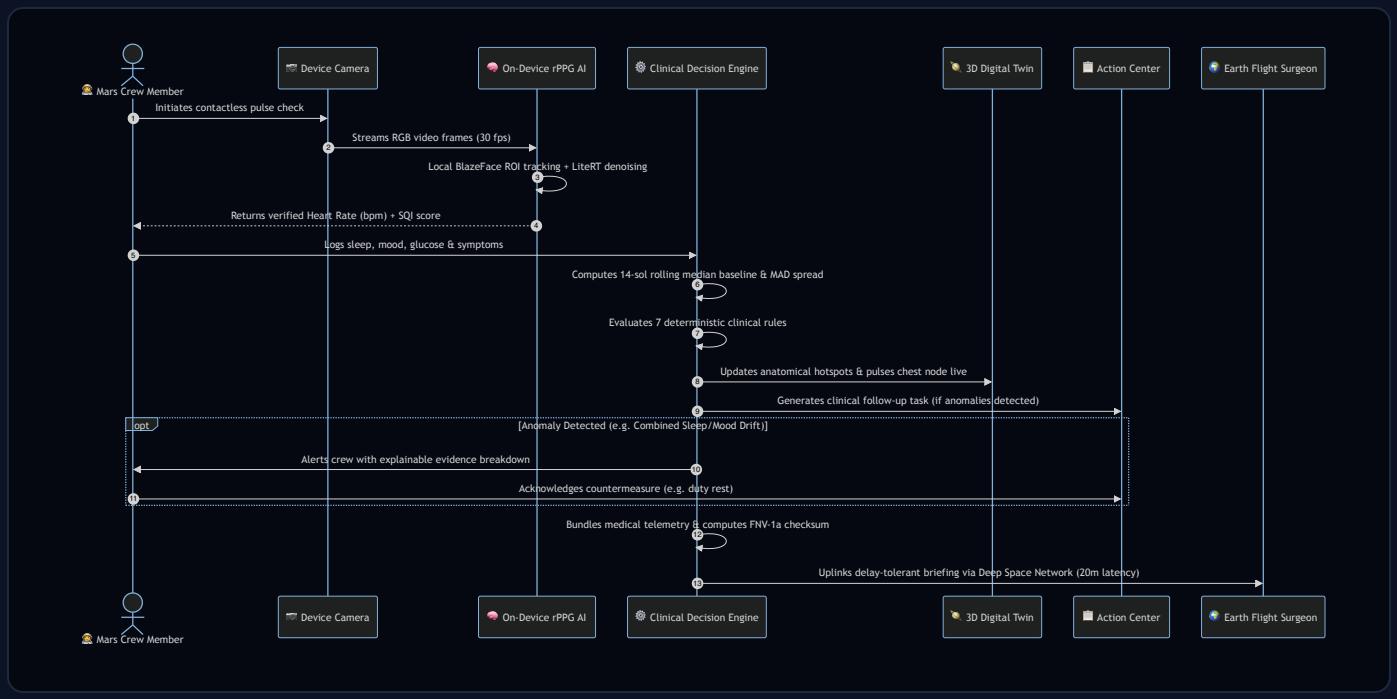
# 1. Master System Architecture Flowchart

## Diagram 1.1: End-to-End 5-Layer System Architecture & Telemetry Pipeline



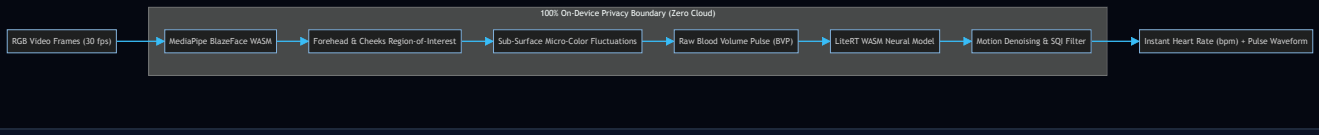
# 2. Astronaut Daily Operational Routine Sequence

## Diagram 2.1: Sol 43 Daily Autonomous Check-in & Uplink Sequence



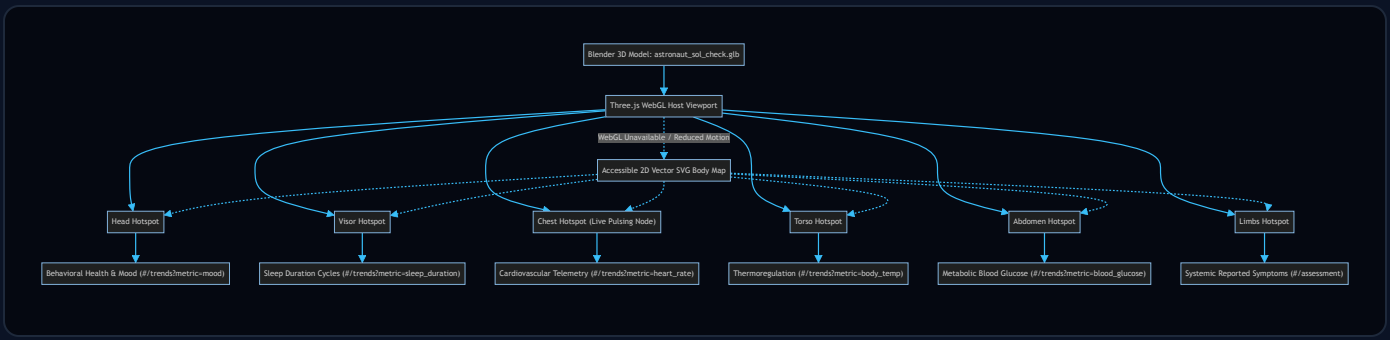

# 3. Contactless Facial rPPG Optical AI Pipeline

## Diagram 3.1: Air-Gapped WebAssembly rPPG Biometric Extraction



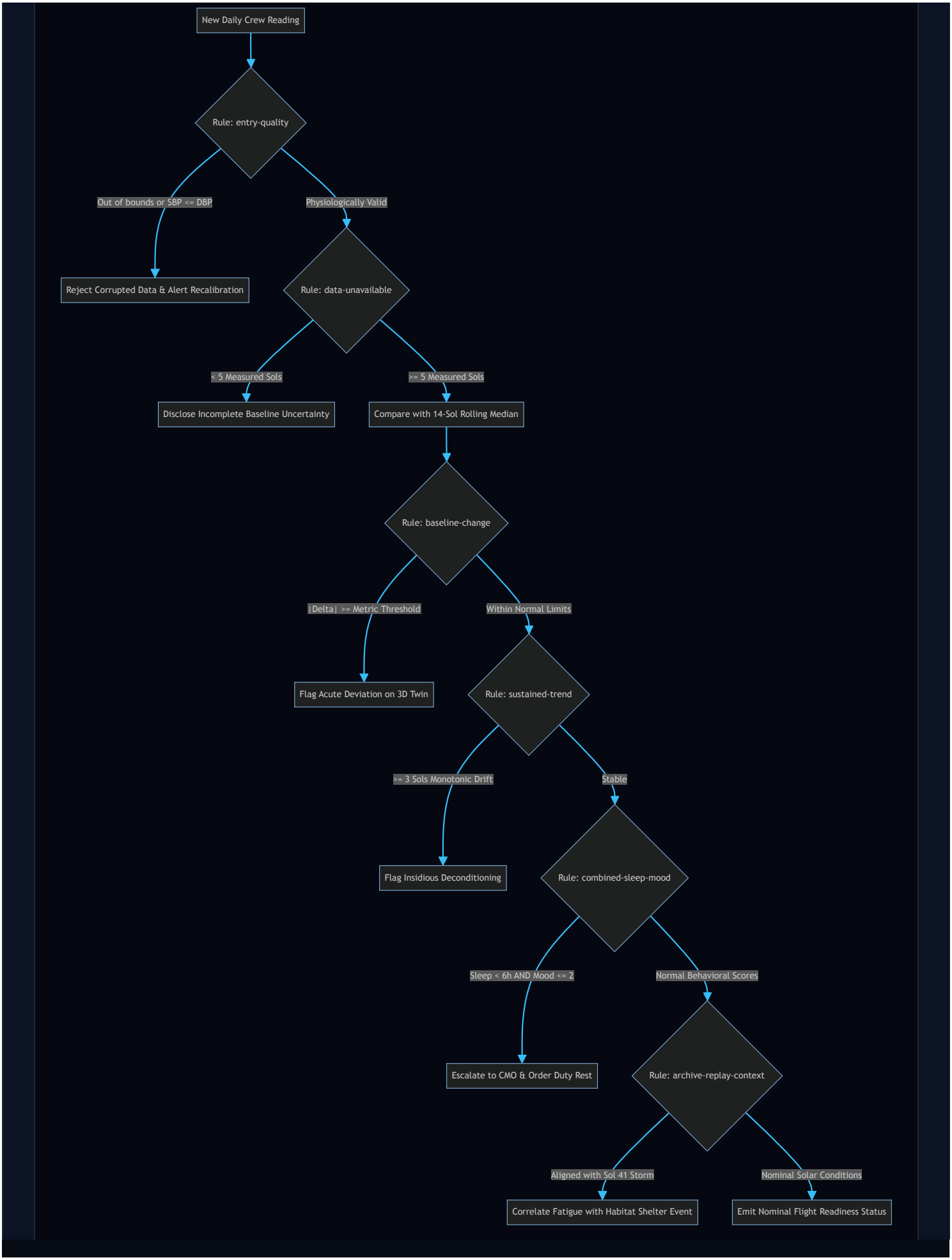
# 4. 3D Astronaut Digital Twin Interactive Mapping

## Diagram 4.1: Blender Model Hotspots & Raycasted Anatomical Domains



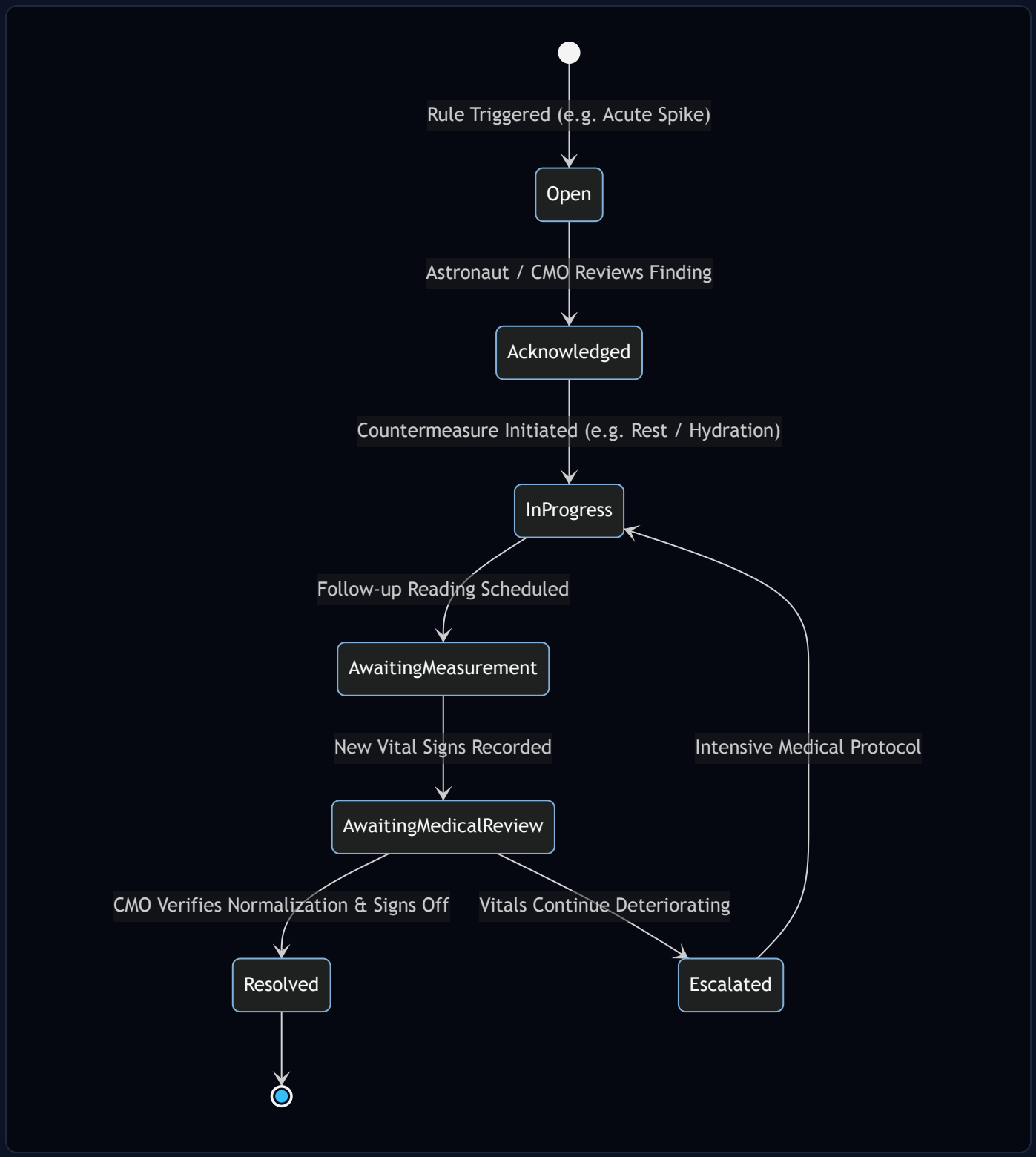
## 5. Deterministic Clinical Heuristics & Decision Tree

### Diagram 5.1: 7-Rule Deterministic Evaluation Flow (No LLM Inference)


New Daily Crew Reading
Rule: entry-quality
Out of bounds or SBP <= DBP
Physiologically Valid
Reject Corrupted Data & Alert Recalibration
Rule: data-unavailable
< 5 Measured Sols
>= 5 Measured Sols
Disclose Incomplete Baseline Uncertainty
Compare with 14-Sol Rolling Median
Rule: baseline-change
|Delta| >= Metric Threshold
Within Normal Limits
Flag Acute Deviation on 3D Twin
Rule: sustained-trend
>= 3 Sols Monotonic Drift
Stable
Flag Insidious Deconditioning
Rule: combined-sleep-mood
Sleep < 6h AND Mood <= 2
Normal Behavioral Scores
Escalate to CMO & Order Duty Rest
Rule: archive-replay-context
Aligned with Sol 41 Storm
Nominal Solar Conditions
Correlate Fatigue with Habitat Shelter Event
Emit Nominal Flight Readiness Status

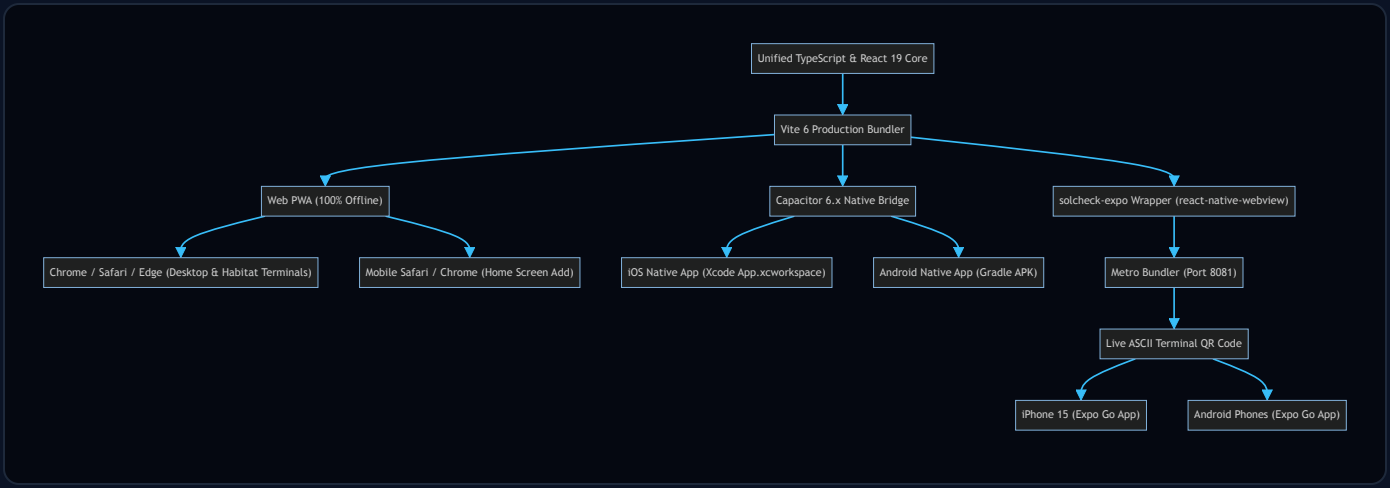
# 6. Closed-Loop Clinical Action State Machine

### Diagram 6.1: Medical Countermeasure Lifecycle & Triage States



# 7. Multi-Platform Cross-Deployment Topology

## Diagram 7.1: Universal Codebase Target Matrix (Web, iOS, Android, Expo Go)



## 8. Explainable Decision Rules Reference Table

| Rule ID | Rule Name | Clinical Trigger Condition | Operational Action |
|---|---|---|---|
| `entry-quality` | Physiological Bounds | Values outside survivable human limits or SBP ≤ DBP | Rejects corrupted sensor data; orders recalibration |
| `combined-sleep-mood` | Behavioral Cascade | Sleep < 6.0h AND Mood ≤ 2 across ≥ 3 consecutive sols | Escalates to CMO; orders sleep hygiene & duty rest |
| `sustained-trend` | Longitudinal Drift | ≥ 3 consecutive readings shifting monotonically | Flags insidious deconditioning (e.g. fluid shifts) |
| `baseline-change` | Acute Anomaly | Single reading deviates by ≥ metric demo delta | Highlights 3D Twin hotspot; emits clinical warning |
| `reported-symptoms` | Symptom Recurrence | Active reported symptoms (e.g. cephalalgia, fatigue) | Correlates symptoms with active mission hazards |
| `archive-replay-context` | Radiation Event Nexus | Correlated with Sol 41 Curiosity RAD storm (>15 µGy/h) | Contextualizes crew fatigue with habitat storm shelter |
| `data-unavailable` | Evidence Disclosure | Fewer than 5 distinct measured sols recorded | Discloses uncertainty; prevents premature triage |

## 9. Quality Assurance & Testing Suite

**Vitest Unit & Integration Suites:** 21 passed (21 total), 124 tests passed (124 total) in ~500ms.

**Chrome DevTools Protocol (CDP) Offline Verification:** 100% route traversal passed with simulated zero-network connectivity.

## 10. Team Roster & Attributions

**Crew Engineers & Authors:** Ahmed Shacker, Ahmed Sherif, Andrew Sameh, David Michael, Hassan Mohsen, Ziad AbdElhaleem.

**Coach:** Mariam Mohamed | **Local Lead:** Dr. Moatasem Mahran.

**3D Model:** Original astronaut 3D model designed and authored by the team in Blender.